

Measure Theory – 2

σ -Algebras and Partitions, Additive Set Functions,
Lebesgue–Stieltjes Measures, $\limsup$ / $\liminf$ of Events,
and Uniqueness of Probability Measures

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1 Sigma-Algebras and Partitions

1.1 Definition and Basic Properties

Definition 1.1 (σ -algebra). Let Ω be a non-empty set. A collection $\mathcal{F}$ of subsets of Ω is a **σ -algebra** (or σ -field) on Ω if

- (i) $\Omega \in \mathcal{F}$ (contains the whole space),
- (ii) $A \in \mathcal{F} \implies A^c \in \mathcal{F}$ (closed under complementation),
- (iii) $A_1, A_2, \dots \in \mathcal{F} \implies \bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$ (closed under countable unions).

The pair $(\Omega, \mathcal{F})$ is called a **measurable space**, and sets in $\mathcal{F}$ are called **measurable**.

Proposition 1.2 (Immediate consequences). *If $\mathcal{F}$ is a σ -algebra on Ω , then*

- (a) $\emptyset \in \mathcal{F}$;
- (b) $\mathcal{F}$ is closed under finite unions;
- (c) $\mathcal{F}$ is closed under countable intersections and finite intersections;
- (d) $A, B \in \mathcal{F} \implies A \setminus B = A \cap B^c \in \mathcal{F}$ and $A \triangle B \in \mathcal{F}$;
- (e) $\limsup A_n$ and $\liminf A_n$ belong to $\mathcal{F}$ whenever every $A_n \in \mathcal{F}$.

Proof. (a) $\emptyset = \Omega^c$. (b) Take $A_i = \emptyset$ for $i > m$ in (iii). (c) By De Morgan, $\bigcap_i A_i = (\bigcup_i A_i^c)^c$. (d) Combine (c) with complements; $A \triangle B = (A \setminus B) \cup (B \setminus A)$. (e) Countable unions and intersections of members of $\mathcal{F}$ stay in $\mathcal{F}$, and both sets are built from such operations (see Section 4). $\square$

Example 1.3 (Standard examples). • **Trivial:** $\mathcal{F} = \{\emptyset, \Omega\}$ (smallest σ -algebra).

- **Power set:** $\mathcal{F} = 2^{\Omega}$ (largest σ -algebra).
- **Generated by one set:** $\sigma(\{A\}) = \{\emptyset, A, A^c, \Omega\}$.
- **Borel:** $\mathcal{B}(\mathbb{R}) = \sigma(\{(a, b] : a < b\})$, which also equals the σ -algebra generated by open sets, by closed sets, or by half-lines $(-\infty, x]$.

- **Non-example:** the collection of finite subsets of $\mathbb{N}$ together with their complements is closed under finite unions but *not* countable unions (the set of all even numbers is a countable union of finite sets yet is neither finite nor co-finite).

Definition 1.4 (Generated σ -algebra). For any collection $\mathcal{C}$ of subsets of Ω ,

$$\sigma(\mathcal{C}) = \bigcap \{ \mathcal{G} : \mathcal{G} \text{ is a } \sigma\text{-algebra on } \Omega, \mathcal{C} \subseteq \mathcal{G} \}.$$

An intersection of σ -algebras is again a σ -algebra, so this is well defined and is the smallest σ -algebra containing $\mathcal{C}$.

1.2 Partitions and the σ -Algebra They Generate

Let $\{A_i\}_{i \in I}$ be a **partition** of Ω , i.e.

- $A_i \neq \emptyset$ for all $i \in I$,
- $A_i \cap A_j = \emptyset$ for $i \neq j$,
- $\bigcup_{i \in I} A_i = \Omega$.

Key Result.

The collection of all unions over subfamilies,

$$\mathcal{F} = \left\{ \bigcup_{i \in J} A_i : J \subseteq I \right\},$$

is a σ -algebra on Ω . (By convention the empty union is $\emptyset$, corresponding to $J = \emptyset$.)

Proof. We verify the three axioms.

1. **Entire space.** Taking $J = I$ gives $\bigcup_{i \in I} A_i = \Omega \in \mathcal{F}$.
2. **Complements.** Let $A = \bigcup_{i \in J} A_i$. Because the A_i are pairwise disjoint and cover Ω ,

$$A^c = \bigcup_{i \in I \setminus J} A_i \in \mathcal{F},$$

since $I \setminus J \subseteq I$.

3. **Countable unions.** Let $A^{(k)} = \bigcup_{i \in J_k} A_i \in \mathcal{F}$ for $k = 1, 2, \dots$. Then

$$\bigcup_{k=1}^{\infty} A^{(k)} = \bigcup_{i \in \bigcup_k J_k} A_i \in \mathcal{F},$$

since $\bigcup_k J_k \subseteq I$. □

Remark. • The map $J \mapsto \bigcup_{i \in J} A_i$ is a bijection from 2^I onto $\mathcal{F}$ (it is injective because the A_i are non-empty and disjoint). Hence $|\mathcal{F}| = 2^{|I|}$ when I is finite.

- The sets A_i are the **atoms** of $\mathcal{F}$: no non-empty proper subset of A_i belongs to $\mathcal{F}$.

- If I is countable, then $\mathcal{F} = \sigma(\{A_i\}_{i \in I})$. If I is uncountable, $\mathcal{F}$ is still a σ -algebra (as proven above), but it can be strictly larger than the σ -algebra generated by the A_i (which consists of countable/co-countable unions), so care is needed.

Example 1.5. Let $\Omega = \{1, 2, 3, 4\}$ with partition $A_1 = \{1, 2\}$, $A_2 = \{3\}$, $A_3 = \{4\}$. Then $|\mathcal{F}| = 2^3 = 8$ and

$$\mathcal{F} = \{\emptyset, \{1, 2\}, \{3\}, \{4\}, \{1, 2, 3\}, \{1, 2, 4\}, \{3, 4\}, \Omega\}.$$

The set $\{1\}$ is *not* in $\mathcal{F}$: $\mathcal{F}$ cannot separate 1 from 2.

Remark. • Closure under *finite* unions alone gives only an *algebra* (field), not a σ -algebra.

- A union of two σ -algebras is generally **not** a σ -algebra; an intersection always is.

2 Additive Set Functions

2.1 Definitions

Definition 2.1 (Additive / σ -additive set functions). Let $\mathcal{A}$ be a collection of subsets of Ω and $\nu : \mathcal{A} \rightarrow \mathbb{R}$.

- ν is **(finitely) additive** if for any finite family of disjoint sets $A_1, \dots, A_m \in \mathcal{A}$ with $\bigcup_{i=1}^m A_i \in \mathcal{A}$,

$$\nu\left(\bigcup_{i=1}^m A_i\right) = \sum_{i=1}^m \nu(A_i).$$

- ν is **σ -additive** (countably additive) if the same holds for countable disjoint families $A_1, A_2, \dots$ with $\bigcup_i A_i \in \mathcal{A}$: $\nu(\bigcup_{i=1}^\infty A_i) = \sum_{i=1}^\infty \nu(A_i)$.
- A **measure** is a σ -additive $\mu : \mathcal{F} \rightarrow [0, \infty]$ with $\mu(\emptyset) = 0$; a **probability measure** additionally has $\mu(\Omega) = 1$.

Proposition 2.2 (Properties of a (finite) measure). Let μ be a measure on $(\Omega, \mathcal{F})$ and $A, B, A_n \in \mathcal{F}$.

- Monotonicity:** $A \subseteq B \implies \mu(A) \leq \mu(B)$.
- Subtractivity:** if $A \subseteq B$ and $\mu(A) < \infty$, then $\mu(B \setminus A) = \mu(B) - \mu(A)$.
- Countable subadditivity:** $\mu(\bigcup_n A_n) \leq \sum_n \mu(A_n)$.
- Continuity from below:** $A_n \uparrow A \implies \mu(A_n) \uparrow \mu(A)$.
- Continuity from above:** $A_n \downarrow A$ and $\mu(A_1) < \infty \implies \mu(A_n) \downarrow \mu(A)$.
- Inclusion–exclusion (two sets):** $\mu(A \cup B) + \mu(A \cap B) = \mu(A) + \mu(B)$ when the measures are finite.

2.2 Problem: Conditions for Additivity on $\{1, 2, 3\}$

Let $\Omega = \{1, 2, 3\}$ and let ν be defined on 2^Ω with $\nu(\emptyset) = 0$ and $\nu(\{1, 2, 3\}) = 1$. Write

$$x = \nu(\{1, 2\}), \quad y = \nu(\{2, 3\}), \quad z = \nu(\{1, 3\}).$$

Necessity. If ν is additive, put $p = \nu(\{1\})$, $q = \nu(\{2\})$, $r = \nu(\{3\})$. Then

$$x = p + q, \quad y = q + r, \quad z = p + r, \quad p + q + r = \nu(\Omega) = 1.$$

Summing the first three,

$$x + y + z = 2(p + q + r) = 2.$$

Sufficiency. Conversely, suppose $x + y + z = 2$. Define

$$p = 1 - y, \quad q = 1 - z, \quad r = 1 - x.$$

Then $p + q + r = 3 - (x + y + z) = 1$, and

$$p + q = 2 - y - z = x, \quad q + r = 2 - z - x = y, \quad p + r = 2 - x - y = z.$$

So every two-point set has value equal to the sum of its singletons, and $\nu(\{1\}) + \nu(\{2, 3\}) = p + y = 1 = \nu(\Omega)$ (similarly for the other two splittings), so ν is additive on all of 2^Ω .

Result.

For real-valued ν on $2^{\{1,2,3\}}$ with $\nu(\emptyset) = 0$, $\nu(\Omega) = 1$:

$$\nu \text{ is additive} \iff x + y + z = 2.$$

Moreover, ν is a (non-negative) probability measure if and only if additionally $0 \leq x, y, z \leq 1$, because $p, q, r \geq 0 \iff y, z, x \leq 1$ (and $x = p + q \geq 0$ etc. then follow automatically).

Example 2.3. $(x, y, z) = (\frac{2}{3}, \frac{2}{3}, \frac{2}{3})$ satisfies $x + y + z = 2$: it is the uniform measure with $p = q = r = \frac{1}{3}$. In contrast $(x, y, z) = (1.5, 0.5, 0)$ also sums to 2 and is additive, but gives $p = 0.5$, $q = 1$, $r = -0.5 < 0$, a *signed* additive function, not a probability measure.

Remark. On a finite Ω with $\mathcal{F} = 2^\Omega$, finite additivity and σ -additivity coincide, since a countable disjoint union of non-empty sets must be finite. The distinction matters only on infinite spaces.

3 Induced Measures and Distribution Functions

3.1 Lebesgue–Stieltjes Measures

Definition 3.1 (Distribution function). A function $F : \mathbb{R} \rightarrow \mathbb{R}$ is a **(generalized) distribution function** if it is non-decreasing and right-continuous. It induces a unique measure μ_F on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ satisfying $\mu_F((a, b]) = F(b) - F(a)$ for all $a < b$. (For a probability distribution, additionally $F(-\infty) = 0$, $F(+\infty) = 1$.)

Evaluating μ_F on intervals.

For $a < b$, writing $F(x^-) = \lim_{t \uparrow x} F(t)$:

$$\begin{aligned} \mu_F((a, b]) &= F(b) - F(a), & \mu_F((a, b)) &= F(b^-) - F(a), \\ \mu_F([a, b]) &= F(b) - F(a^-), & \mu_F([a, b)) &= F(b^-) - F(a^-), \\ \mu_F(\{p\}) &= F(p) - F(p^-). \end{aligned}$$

μ_F has an **atom** at p exactly when F jumps at p , and the atom's mass equals the size of the jump.

3.2 Problem: A Piecewise Distribution Function

Consider

$$F(x) = \begin{cases} 1+x, & -15 \leq x < 0, \\ 2+x^2, & 0 \leq x < 2, \\ 9, & x \geq 2. \end{cases}$$

Key values.

x	-15	-1	0^-	0	1	2^-	
$F(x)$	-14	0	1	2	3	6	and $F(2) = 9$.

F is non-decreasing and right-continuous ($F(0) = 2 = \lim_{x \downarrow 0} (2 + x^2)$), with jumps at $x = 0$ (size $2 - 1 = 1$) and at $x = 2$ (size $9 - 6 = 3$).

(a) Singleton $\{2\}$.

$$\mu_F(\{2\}) = F(2) - F(2^-) = 9 - (2 + 2^2) = 9 - 6 = 3.$$

(b) The set $(-1, 0] \cup (1, 2)$. The intervals are disjoint, so their measures add:

$$\mu_F((-1, 0]) = F(0) - F(-1) = (2 + 0^2) - (1 + (-1)) = 2 - 0 = 2,$$

$$\mu_F((1, 2)) = F(2^-) - F(1) = 6 - (2 + 1^2) = 6 - 3 = 3.$$

Therefore

$$\mu_F((-1, 0] \cup (1, 2)) = 2 + 3 = 5.$$

Structure of μ_F and a consistency check. On $(-15, \infty)$, μ_F decomposes as

$$\mu_F = \underbrace{\lambda|_{(-15,0)}}_{\text{density 1}} + \underbrace{1 \cdot \delta_0}_{\text{atom at 0}} + \underbrace{2x \lambda|_{(0,2)}}_{\text{density } 2x} + \underbrace{3 \cdot \delta_2}_{\text{atom at 2}},$$

where λ is Lebesgue measure. This agrees with the computations above:

- $\mu_F((-1, 0]) = \underbrace{1}_{\text{mass on } (-1,0)} + \underbrace{1}_{\text{atom at 0}} = 2;$
- $\mu_F((1, 2)) = \int_1^2 2x \, dx = 4 - 1 = 3;$
- total mass on $(-15, \infty)$: $15 + 1 + 4 + 3 = 23 = F(\infty) - F(-15) = 9 - (-14).$

Remark. • Whether an endpoint is open or closed matters exactly when F jumps there. For example $\mu_F((-1, 0)) = F(0^-) - F(-1) = 1$, while $\mu_F((-1, 0]) = 2$.

- Always use $F(p^-)$ (the *left* limit) for the mass of $\{p\}$; using $F(p^+)$ always yields 0 by right-continuity.

4 Asymptotic Events: Infinitely Often vs. Almost Always

4.1 Definitions

For events $\{A_n\}_{n \geq 1}$ in a probability space $(\Omega, \mathcal{F}, \mathbb{P})$:

$$\begin{aligned}\{A_n \text{ i.o.}\} &= \limsup_{n \rightarrow \infty} A_n = \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k, \\ \{A_n \text{ a.a.}\} &= \liminf_{n \rightarrow \infty} A_n = \bigcup_{n=1}^{\infty} \bigcap_{k=n}^{\infty} A_k.\end{aligned}$$

- $\omega \in \{A_n \text{ i.o.}\}$ iff ω lies in **infinitely many** A_n (for every n there is some $k \geq n$ with $\omega \in A_k$).
- $\omega \in \{A_n \text{ a.a.}\}$ iff ω lies in **all but finitely many** A_n (there is an n such that $\omega \in A_k$ for every $k \geq n$).

4.2 Relationships

1. Inclusion.

$$\{A_n \text{ a.a.}\} \subseteq \{A_n \text{ i.o.}\}.$$

Proof. If ω is in all but finitely many A_n , it is in infinitely many.

2. Duality (De Morgan).

$$\{A_n \text{ a.a.}\}^c = \left(\bigcup_{n=1}^{\infty} \bigcap_{k=n}^{\infty} A_k \right)^c = \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k^c = \{A_n^c \text{ i.o.}\}.$$

Symmetrically, $\{A_n \text{ i.o.}\}^c = \{A_n^c \text{ a.a.}\}.$

3. Indicator characterization.

$\mathbf{1}_{\limsup A_n} = \limsup_n \mathbf{1}_{A_n}$ and $\mathbf{1}_{\liminf A_n} = \liminf_n \mathbf{1}_{A_n}.$

4. Monotone limits.

If $A_n \uparrow A$ or $A_n \downarrow A$, then $\limsup A_n = \liminf A_n = A.$

5. Measurability.

Both sets are countable unions/intersections of events, so both belong to $\mathcal{F}.$

Example 4.1. Let $A_n = A$ for even n and $A_n = A^c$ for odd n . Then $\limsup A_n = A \cup A^c = \Omega$ (every ω is in infinitely many A_n), while $\liminf A_n = A \cap A^c = \emptyset$ (no ω is in all but finitely many). So the inclusion $\liminf \subseteq \limsup$ can be strict.

4.3 Probability of $\limsup$ and $\liminf$

Proposition 4.2 (Fatou for sets).

$$\mathbb{P}(\liminf A_n) \leq \liminf_n \mathbb{P}(A_n) \leq \limsup_n \mathbb{P}(A_n) \leq \mathbb{P}(\limsup A_n).$$

Proof. Let $B_n = \bigcap_{k \geq n} A_k \uparrow \liminf A_n$. Since $B_n \subseteq A_n$, $\mathbb{P}(B_n) \leq \mathbb{P}(A_n)$, and by continuity from below $\mathbb{P}(\liminf A_n) = \lim_n \mathbb{P}(B_n) \leq \liminf_n \mathbb{P}(A_n)$. The last inequality is the same argument applied to $C_n = \bigcup_{k \geq n} A_k \downarrow \limsup A_n$, using continuity from above and $A_n \subseteq C_n$. $\square$

Theorem 4.3 (First Borel–Cantelli lemma). *If $\sum_{n=1}^{\infty} \mathbb{P}(A_n) < \infty$, then $\mathbb{P}(A_n \text{ i.o.}) = 0$.*

Proof. For every n , $\limsup A_k \subseteq \bigcup_{k \geq n} A_k$, so by subadditivity $\mathbb{P}(A_n \text{ i.o.}) \leq \sum_{k \geq n} \mathbb{P}(A_k)$. The right side is the tail of a convergent series and tends to 0 as $n \rightarrow \infty$. $\square$

Theorem 4.4 (Second Borel–Cantelli lemma). *If the events A_n are independent and $\sum_{n=1}^{\infty} \mathbb{P}(A_n) = \infty$, then $\mathbb{P}(A_n \text{ i.o.}) = 1$.*

Proof. It suffices to show $\mathbb{P}(\{A_n \text{ i.o.}\}^c) = \mathbb{P}(A_n^c \text{ a.a.}) = 0$. For fixed n and $m \geq n$, independence and $1 - t \leq e^{-t}$ give

$$\mathbb{P}\left(\bigcap_{k=n}^m A_k^c\right) = \prod_{k=n}^m (1 - \mathbb{P}(A_k)) \leq \exp\left(-\sum_{k=n}^m \mathbb{P}(A_k)\right) \xrightarrow{m \rightarrow \infty} 0.$$

Hence $\mathbb{P}\left(\bigcap_{k \geq n} A_k^c\right) = 0$ for each n , and the countable union over n also has probability 0. $\square$

Remark. Together, the Borel–Cantelli lemmas give a zero–one law for independent events: $\mathbb{P}(A_n \text{ i.o.}) \in \{0, 1\}$, according to whether $\sum \mathbb{P}(A_n)$ converges or diverges. Independence cannot be dropped from the second lemma: take $A_n = A$ for all n with $\mathbb{P}(A) = \frac{1}{2}$; then $\sum \mathbb{P}(A_n) = \infty$ but $\mathbb{P}(A_n \text{ i.o.}) = \frac{1}{2}$.

5 Uniqueness and Equivalence of Probability Measures

5.1 Problem: Agreement on “Small” Sets

Suppose $\mathbb{P}$ and Q are probability measures on $(\Omega, \mathcal{F})$ such that

$$\mathbb{P}(A) = Q(A) \quad \text{for all } A \in \mathcal{F} \text{ with } \mathbb{P}(A) \leq \frac{1}{2}.$$

Claim: $\mathbb{P} = Q$ on $\mathcal{F}$.

Proof. Let $A \in \mathcal{F}$.

- **Case 1:** $\mathbb{P}(A) \leq \frac{1}{2}$. Then $\mathbb{P}(A) = Q(A)$ by hypothesis.
- **Case 2:** $\mathbb{P}(A) > \frac{1}{2}$. Consider $A^c \in \mathcal{F}$. Then

$$\mathbb{P}(A^c) = 1 - \mathbb{P}(A) < 1 - \frac{1}{2} = \frac{1}{2},$$

so Case 1 applies to A^c : $\mathbb{P}(A^c) = Q(A^c)$. Therefore

$$Q(A) = 1 - Q(A^c) = 1 - \mathbb{P}(A^c) = \mathbb{P}(A).$$

In both cases $\mathbb{P}(A) = Q(A)$, so $\mathbb{P} = Q$ on all of $\mathcal{F}$. $\square$

Remark. The proof uses only (i) $\mathcal{F}$ is closed under complements and (ii) both measures are normalized ($\mathbb{P}(\Omega) = Q(\Omega) = 1$). The threshold $\frac{1}{2}$ is essential: the hypothesis must cover every A or its complement. With a threshold $c < \frac{1}{2}$, sets with $\mathbb{P}(A) \in (c, 1 - c)$ would be left unconstrained.

5.2 Related Uniqueness Tools

Theorem 5.1 (π - λ uniqueness theorem). *Let $\mathcal{P}$ be a π -system (closed under finite intersections) with $\sigma(\mathcal{P}) = \mathcal{F}$. If two probability measures agree on $\mathcal{P}$, they agree on $\mathcal{F}$.*

Example 5.2. Since $\{(-\infty, x] : x \in \mathbb{R}\}$ is a π -system generating $\mathcal{B}(\mathbb{R})$, a probability measure on $\mathbb{R}$ is determined by its distribution function F . This is why Lebesgue–Stieltjes measures (Section 3) are well defined.

References

- [1] S. Basu, T. Butler, D. Estep, and N. Panda, *A Ramble Through Probability: How I Learned to Stop Worrying and Love Measure Theory*, Computational Science & Engineering, vol. 29, SIAM, Philadelphia, 2024.